

Morphic Studio

Document 25 — Rendering & Export System (Expanded Specification)

Document ID: MS-025

Version: 2.0

Status: Expanded Specification

Priority: Core Infrastructure

Purpose

The Rendering & Export System transforms Morphic Studio projects into finished deliverables.

It is responsible for rendering comics, motion comics, animations, storyboards, project packages, and reusable assets while maintaining quality, consistency, and creator intent.

The system supports multiple output formats without requiring creators to rebuild their projects.

Vision

A creator should be able to build one project and publish it everywhere.

Instead of recreating work for different platforms, Morphic Studio adapts the project to the chosen output format.

Rendering should be reliable, scalable, and preserve artistic quality.

Core Philosophy

Create Once. Publish Anywhere.

Every project should remain platform-independent until export.

The export system adapts the content while preserving the original project.

Objectives

The Rendering & Export System shall:

- Render comics.
 - Render motion comics.
 - Render animations.
 - Export reusable assets.
 - Support multiple quality levels.
 - Support multiple resolutions.
 - Support background rendering.
 - Preserve project integrity.
 - Automate export workflows.
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Rendering Pipeline

Every export follows:

Project

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Validation

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Asset Collection

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Missing Asset Check

↓

Render Planning

↓

Rendering



Quality Verification



Compression (Optional)



Packaging



Export



Verification



Delivery

Supported Export Categories

Comics

- PDF
 - PNG
 - JPEG
 - TIFF
 - Print Package
 - Webtoon Package
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Motion Comics

- MP4
 - MOV
 - Web Package
 - Interactive Package (Future)
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Animation

- MP4
 - MOV
 - Image Sequence
 - Animated GIF
 - Alpha Channel Video (Future)
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Storyboards

- PDF
 - Image Sequence
 - Presentation Package
-

Assets

- Character Package
 - Background Package
 - Prop Package
 - Voice Package
 - Animation Package
-

Project

- Full Project Backup
 - Archive
 - Transfer Package
 - Template Package
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Rendering Profiles

Creators may choose:

Draft



Standard



High Quality



Ultra



Custom

Each profile controls:

- Resolution
- Compression
- Sampling
- Effects
- Render Priority
- Output Size

Resolution Support

Supported resolutions include:

- 720p
- 1080p
- 1440p
- 4K
- 8K (Future)
- Custom

For images:

- Web
- Print
- Poster
- Creator-defined dimensions

Batch Rendering

Creators may render:

- Entire project

- Selected chapters
- Selected scenes
- Individual pages
- Individual panels
- Individual animation shots

Batch jobs execute through the Workflow Engine.

Background Rendering

Rendering occurs in the background whenever possible.

Creators may continue:

- Writing
- Editing
- Storyboarding
- Planning

while rendering proceeds independently.

Quality Validation

Before export, the system checks:

- Missing assets
- Broken references
- Missing dialogue
- Audio synchronization
- Subtitle timing
- Resolution mismatches
- Rendering errors

Warnings are presented before final export.

Compression

Creators may choose:

- Maximum Quality
- Balanced
- Small File Size

- Custom Settings

Compression should never modify original project data.

Export Templates

Reusable templates include:

- Manga Print
- American Comic
- Webtoon
- Instagram
- TikTok Preview
- YouTube Preview
- Presentation
- Portfolio

Creators may create custom templates.

User Stories

As a creator, I want to export my project in multiple formats without rebuilding it.

As a creator, I want rendering to happen in the background.

As a creator, I want reusable export presets.

As a creator, I want confidence that my exported project matches my approved production.

Functional Requirements

The Rendering & Export System shall allow users to:

- Render projects.
- Pause rendering.
- Resume rendering.
- Cancel rendering.
- Save export presets.
- Preview exports.
- Batch export.
- Schedule rendering.

- Validate output.
 - Archive completed renders.
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UI Components

The interface includes:

- Render Queue
 - Export Manager
 - Export Presets
 - Render Progress
 - Quality Settings
 - Resolution Selector
 - Compression Settings
 - Preview Window
 - Render History
 - Delivery Manager
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Database Entities

Core entities include:

- RenderJob
 - ExportProfile
 - RenderQueue
 - OutputPackage
 - RenderHistory
 - DeliveryRecord
 - ValidationReport
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API Considerations

Potential endpoints include:

- Create Render Job
- Retrieve Render Status
- Pause Render
- Resume Render
- Cancel Render
- Export Project
- Retrieve Export History

- Validate Export
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Non-Functional Requirements

The Rendering & Export System should:

- Support long-running render jobs.
 - Resume interrupted renders.
 - Handle high-resolution projects.
 - Scale to large productions.
 - Integrate with distributed rendering in the future.
 - Preserve original assets.
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Error Handling

If rendering fails:

- Preserve completed work.
- Resume from the last successful checkpoint.
- Explain the issue.
- Retry failed stages where appropriate.

If required assets are unavailable:

- Pause rendering.
 - Notify the creator.
 - Resume after resolution.
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Future Expansion

Future versions may include:

- Distributed cloud rendering.
 - GPU render farms.
 - Real-time previews.
 - Adaptive compression.
 - Platform-specific optimization.
 - One-click publishing.
 - Live streaming exports.
 - Interactive web exports.
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Acceptance Criteria

The Rendering & Export System succeeds when creators can reliably transform any Morphic Studio project into high-quality deliverables for multiple platforms while preserving creative intent, production quality, and project integrity.

Key Principle

A finished project should not be limited by its destination. Morphic Studio separates creation from delivery, allowing creators to build once and publish anywhere with confidence.

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